Introduction to Statistical Methods(ISM)

Course No: AIML ZC418
Course Title: ISM
WEBINAR 3 : 13/01/2026

BITS Pilani

IMP Note to Self

Work Integrated Learning Programmes

Confidence interval for Mean

A **100(1 − α)% confidence interval** for the mean μ of a normal population when the value of σ is known is given by

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) \tag{7.5}$$

or, equivalently, by $\bar{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$.

If n is sufficiently large, the standardized variable

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has approximately a standard normal distribution. This implies that

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \tag{7.8}$$

is a **large-sample confidence interval for μ** with confidence level approximately 100(1 − α)%. This formula is valid regardless of the shape of the population distribution.

Work Integrated Learning Programmes

Confidence interval :

Let $\bar{x}$ and s be the sample mean and sample standard deviation computed from the results of a random sample from a normal population with mean μ. Then a **$100(1-\alpha)\%$ confidence interval for μ** is

$$\left(\bar{x} - t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}},\ \bar{x} + t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}\right) \tag{7.15}$$

or, more compactly, $\bar{x} \pm t_{\alpha/2,n-1} \cdot s/\sqrt{n}$.

Formula for a Specific Confidence Interval for a Proportion

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

when $n\hat{p}$ and $n\hat{q}$ are each greater than or equal to 5.

Symbols Used in Proportion Notation

p = population proportion

$\hat{p}$ (read "p hat") = sample proportion

For a sample proportion,

$$\hat{p} = \frac{X}{n} \quad \text{and} \quad \hat{q} = \frac{n-X}{n} \quad \text{or} \quad \hat{q} = 1 - \hat{p}$$

where X = number of sample units that possess the characteristics of interest and n = sample size.

Problem 1:

Assume that the helium porosity (in percentage) of coal samples taken from any particular seam is normally distributed with true standard deviation .75.
Compute a 98% CI for the true average porosity of a certain seam if the average porosity for 16 specimens from the seam was 4.56.
Solution:

$$n = 16;\ \bar{X} = 4.56\ ;\ \sigma = 0.75\ ; 1 - \alpha = 0.98 \Longrightarrow \alpha = 0.02; Z_{\frac{\alpha}{2}} = 2.33$$

Confidence interval for mean is $\bar{X} - Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$$4.56 - 2.33 \times \frac{0.75}{\sqrt{16}} < \mu < 4.56 + 2.33 \times \frac{0.75}{\sqrt{16}}$$

$$4.56 - 0.437 < \mu < 4.56 + 0.437$$

$$4.123 < \mu < 4.997$$

Problem 2:

Find the 98% confidence interval for the average time students spend on a deep learning project. A sample of 15 students was taken, and their recorded project completion times (in hours) were: 76, 85, 90, 78, 83, 88, 91, 75, 84, 79, 87, 92, 80, 86, 89 Assume the time spent follows a normal distribution, but the population standard deviation is unknown.

Solution:

n=15 ;$\bar{X} = 84.2\,; s = 5.52\,; 1-\alpha = 0.98 \Longrightarrow \alpha = 0.02; \mathrm{t}_{\frac{\alpha}{2}} = 2.624$

Confidence interval for mean is

$$\bar{X} - \mathrm{t}_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} < \mu < \bar{X} + \mathrm{t}_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$84.2 - 2.624 \times \frac{5.52}{\sqrt{15}} < \mu < 84.2 + 2.624 \times \frac{5.52}{\sqrt{15}}$$

$$80.46 < \mu < 87.94$$

Sample standard deviation calculation:

76, 85, 90, 78, 83, 88, 91, 75, 84, 79, 87, 92, 80, 86,

$s^2 = 30.46 \Rightarrow s = 5.52$

x	$x - \bar{x}$	$(x - \bar{x})^2$
76	-8.2	67.24
85	0.8	0.64
90	5.8	33.64
78	-6.2	38.44
83	-1.2	1.44
88	3.8	14.44
91	6.8	46.24
75	-9.2	84.64
84	-0.2	0.04
79	-5.2	27.04
87	2.8	7.84
92	7.8	60.84
80	-4.2	17.64
86	1.8	3.24
89	4.8	23.04
$\bar{x} = 84.2$		$\sum (x - \bar{x})^2 = 426.4$ $\frac{1}{n-1}\sum (x - \bar{x})^2 = 30.46$

Problem 3:

A random sample of 400 items is found to have a mean of 82 and standard deviation of 18. Find 99%confidence interval for the mean of the population from which the sample is drawn.

Solution:

$$n = 400;\ \bar{X} = 82\ ; s = 18\ ; 1 - \alpha = 0.99 \Longrightarrow \alpha = 0.01; Z_{\frac{\alpha}{2}} = 2.575$$

Confidence interval for mean is $\bar{X} - Z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \frac{s}{\sqrt{n}}$

$$82 - 2.575 \times \frac{18}{\sqrt{400}} < \mu < 82 + 2.575 \times \frac{18}{\sqrt{400}}$$

$$82 - 2.3175 < \mu < 82 + 2.3175$$

$$79.6825 < \mu < 84.3175$$

Problem 4:

The article "Evaluating Tunnel Kiln Performance" (*Amer.Ceramic Soc. Bull.*, Aug. 1997: 59–63) gave the following summary information for fracture strengths (MPa) of n=169; ceramic bars fired in a particular kiln: $\bar{X} = 89.10; s = 3.73$

a. Calculate a (two-sided) confidence interval for true average fracture strength using a confidence level of 95%.

b. Suppose the investigators had believed *a priori* that the population standard deviation was about 4 MPa. Based on this supposition, how large a sample would have been required to estimate μ to within .5 MPa with 95% confidence?

Solution: a) $n = 169\ ;\ \bar{X} = 89.10\ ; s = 3.73\ ; 1 - \alpha = 0.95 \Longrightarrow \alpha = 0.05; Z_{\frac{\alpha}{2}} = 1.96$

$$\bar{X} - Z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$89.10 - 1.96 \times \frac{3.73}{\sqrt{169}} < \mu < 89.10 + 1.96 \times \frac{3.73}{\sqrt{169}}$$

$$89.10 - 0.5624 < \mu < 89.10 + 0.5624$$

$$88.54 < \mu < 89.66$$

Solution contd.,

Suppose the investigators had believed *a priori* that the population standard deviation was about 4 MPa. Based on this supposition, how large a sample would have been required to estimate μ to within .5 MPa with 95% confidence?

$$b)\ n = ?\ ;\ \bar{X} = 89.10\ ; s = 4\ ; 1 - \alpha = 0.95 \Longrightarrow \alpha = 0.05; Z_{\frac{\alpha}{2}} = 1.96;$$

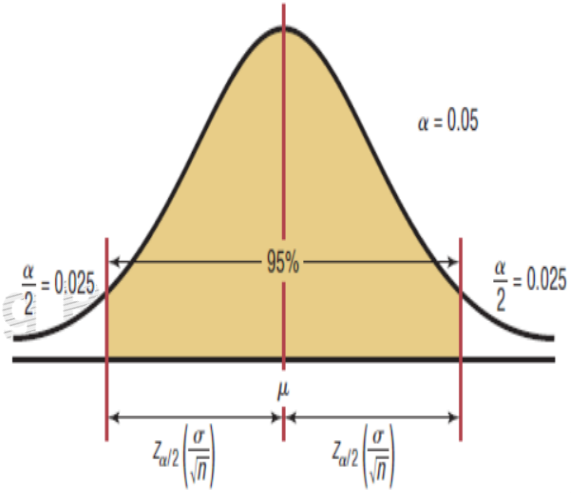
$$margin\ of\ \ error\ , E = 0.5$$

$$E = Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

$$\Longrightarrow 0.5 = 1.96 \times \frac{4}{\sqrt{n}}$$

$$\Longrightarrow \mathrm{n} = \left(1.96 \times \frac{4}{0.5}\right)^2$$

$$= 245.86$$



Required sample size is 246

Statistical Hypothesis

A **statistical hypothesis,** or just *hypothesis,* is a claim or assertion either about the value of a single parameter (population characteristic or characteristic of a probability distribution), about the values of several parameters, or about the form of an entire probability distribution.

There are two types of statistical hypotheses for each situation: the null hypothesis and the alternative hypothesis.

The **null hypothesis,** symbolized by H_0, is a statistical hypothesis that states that there
is no difference between a parameter and a specific value, or that there is no difference
between two parameters.
The **alternative hypothesis,** symbolized by H_1, is a statistical hypothesis that states the
existence of a difference between a parameter and a specific value, or states that there is a difference between two parameters.

exercise

For the following statements (claims) state H_0 , H_1 (or H_a)

1. The average age of community college students is 24.6 years.

   Solution: H_0: $\mu=24.6$ versus H_a: $\mu \neq 24.6$

2. The average score of high school basketball games is less than 88.

   Solution: H_0: $\mu \geq 88$ versus H_a: $\mu < 88$.

3. A random sample of n=35 observations from a quantitative population produced a mean $\bar{x} = 2.4$ and a standard deviation s=0.29. Suppose your research objective is to show that the population mean μ exceeds 2.3. Give the null and alternative hypotheses for the test.

   Solution: H_0: $\mu \leq 2.3$ versus H_a: $\mu > 2.3$.

Approach to Hypothesis Testing with Fixed Probability of Type I Error

1. State the null and alternative hypotheses.

2. Choose a fixed significance level α.

3. Choose an appropriate test statistic and establish the critical region based on α.

4. Reject H_0 if the computed test statistic is in the critical region. Otherwise, do not reject.

5. Draw scientific or engineering conclusions.

Rejection criterion:

$$\boldsymbol{Rejection\ region}:$$
$$|z| > z_{\frac{\alpha}{2}}\ (two - tailed\ RR)$$

$$z < -z_{\alpha}(lower - tail\ RR)$$
$$z > z_{\alpha}\ (upper - tail\ RR)$$

$$\boldsymbol{Rejection\ region}:$$
$$|t| > t_{\frac{\alpha}{2}}\ (two - tailed\ RR)$$
$$t < -t_{\alpha}(lower - tail\ RR)$$

$$t > t_{\alpha}\ (upper - tail\ RR)$$

P-Value Approach

1. **State the hypotheses and identify the claim.**
2. **Compute the test value.**
3. **Find the *P*-value.**
4. **Make the decision to reject or not reject the null hypothesis based on the P-value.**
5. **Summarize the results.**

Decision Rule When Using a *P*-Value

If *P*-value < α, reject the null hypothesis.
If *P*-value $\geq \alpha$, do not reject the null hypothesis.

To find the P-value.

a. For a left-tailed test, P = (Area in left tail).

b. For a right-tailed test, P = (Area in right tail).

c. For a two-tailed test, P = 2(Area in tail of standardized test statistic).

Exercise

Find the P-value for the hypothesis test with the standardized test statistic z. Decide whether to reject H_0 for the level of significance

1. **Left-tailed test z = -1.32 ; α=0.1.**
2. **Right tailed test ,z = 1.23; α=0.05.**
3. **Two tailed test, z=-1.68; α=0.05.**

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379

1) Left-tailed test
z = −1.32, α = 0.10
P-value = $P(Z \leq -1.32) = 0.0934$
Since $0.0934 < 0.10 \rightarrow$ Reject H_0

2) Right-tailed test
z = 1.23, α = 0.05
P-value = $P(Z \geq 1.23) = 1 - P(Z \leq 1.23) = 1 - 0.8907 = 0.1093$
Since $0.1093 > 0.05 \rightarrow$ Fail to reject H_0

3) Two tailed test z = −1.68, α = 0.05
P-value = $2 \times P(Z \leq -1.68) = 2(0.0465) = 0.093$
Since $0.093 > 0.05 \rightarrow$ Fail to reject H_0

Which test ?

Hypothesis Testing	Sample Size	Test Statistic
1) Test for specified mean	Large or small with σ known	$z=\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$
	Small (with (n-1) d.o.f.), σ unknown	$t=\frac{\bar{x}-\mu}{s/\sqrt{n}}$

Remark: If the population standard deviation σ is unknown then we use t-distribution. But whenever sample size is large then t -distribution is nearly equal to Z-distribution.

Critical values of Z

Level of Significance	*Two-Tailed Test*	*One-Tailed Test*
0.10 = 10%	±1.65	+1.28 or −1.28
0.05 = 5%	±1.96	+1.65 or −1.65
0.01 = 1%	±2.58	+2.33 or −2.33

Which test ?

Hypothesis Testing	Sample Size	Test Statistic
2) Test for equality of two means	Large ((n1 + n2)) or small with σ known	$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \cong N(0,1)$
	Small (with (n1 + n2 - 2) d.o.f.), σ unknown Assume Population variances are equal	$t = \frac{\bar{x}_2 - \bar{x}_1}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$

Which test ?

Hypothesis Testing	Sample Size	Test Statistic
3) Test for specified proportion	Large provided that $np_0 \geq 10$ & $n(1-p_0) \geq 10$	$z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$
4) Test for equality of two proportions	Large ((n1 + n2)) In this case, each of $n_1p_1, n_1q_1, n_2p_2, and\ n_2q_2$ must all be 10 or more.	$z = \dfrac{(\hat{p}_1 - \hat{p}_2) - p_1 - p_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$, where $\hat{p} = \frac{x1 + x2}{n1 + n2}$ and $\hat{p}_1 = \frac{x_1}{n_1}$, $\hat{p}_2 = \frac{x_2}{n_2}$ **OR** $Z = \dfrac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{\frac{p_1q_1}{n_1} + \frac{p_2q_2}{n_2}}} \cong N(0,1)$
5) Student's paired t-test (before & after observations of a single population)	Small (with (n-1) d.o.f.)	$t = \dfrac{\bar{d} - \Delta_0}{s_D/\sqrt{n}}$

Problem : 5

A sample of 30 milk carton provides a sample mean of 505 ml. The population standard deviation is believed to be 10 ml. Perform a hypothesis test, at the 0.05 level of significance, population mean 500 ml and to help determine whether the filling process should continue operating or be stopped and corrected. Use p – value also to validate the hypothesis.

Solution: Here, n = 30, $\bar{x} = 505$, $\sigma = 10$

H_0: $\mu=500$ versus H_a: $\mu \neq 500$

The level of significance is 0.05

The z-statistic is $z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} = \frac{505-500}{10/\sqrt{30}} = \frac{5}{1.826} = 2.738$

$\therefore$Calculated z value = 2.738

The p-value =0.0062

Since p- value is less than level of significance (α), we reject null hypothesis and accept H_1.

Conclusion: At the 0.05 level of significance, there is sufficient evidence to conclude that the population mean milk volume is significantly different from 500 ml. So, the filling process should be stopped and corrected as the mean filling volume is not meeting the target 500 ml.

Problem:6

Let μ denote the true average tread life of a certain type of tire. Consider testing H_0: μ=30,000 versus H_a: μ > 30,000 based on a sample of size n=16 from a normal population distribution with σ =1500. If $\bar{x} = 30,960$ and a level α=0.01 test is used, what is the decision?

Solution: Here, n = 16, $\bar{x} = 30,960$, σ = 1500

H_0: μ=30,000 versus H_a: μ > 30,000

The level of significance is 0.05

The z-statistic is $z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} = \frac{30960-30000}{1500/\sqrt{16}} = 960 / 375 = 2.56$

∴calculated z value = 2.56

The critical value at 0.01 LOS is 2.33

Since the calculated value is greater than the critical value, we reject null hypothesis and accept alternative hypothesis.

Conclusion: The true average tread life of a certain type of tire is more than 30,000

Problem:7

Foody an online food ordering and delivery company is aimed to reduce the delivery time to a minimum. The mean delivery time for a random sample of 25 food orders is 13.15 minutes, and standard deviation is 3.1 minutes. Perform a hypothesis test, with α =0.05 level of significance, to determine whether the population mean delivery time meets the service goal of 10 minutes or less is being achieved.

Solution: Here, n = 25, $\bar{x} = 13.15; s = 3.1$

H0: μ≤10 & H1:μ>10

Here it is one tailed t test

The level of significance is 0.05.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{13.15 - 10}{3.1/\sqrt{25}} = 5.0806$$

∴calculated t value = 5.0806

The critical value from table with 24 degrees of freedom and 0.05 LOS is 1.711

Since the calculated value is less than than the critical value, we reject null hypothesis (as it is left tailed).

Conclusion: At the 5% significance level, there is enough evidence to say that the service time is less than 10 minutes)

Problem:8

A sample of 50 lenses used in eyeglasses yields a sample mean thickness of 3.05 mm and a sample standard deviation of .34 mm. The desired true average thickness of such lenses is 3.20 mm. Does the data strongly suggest that the true average thickness of such lenses is something other than what is desired? Test using α =0.05

Solution: Here, n = 50, $\bar{x}$= 3.05, s = 0.34, μ=3.20

H0: μ=3.20

H1:μ≠3.20

Here it is two tailed z test

The level of significance is 0.05. critical value is ±1.96

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{3.05 - 3.20}{0.34/\sqrt{50}} = -0.15 \,/\, 0.62$$

$$= -0.15/0.048 = -3.125$$

∴calculated z value = |-3.125| = 3.125

Since the calculated value is greater than the critical value, we reject null hypothesis.

Conclusion: There is strong evidence to say that the data strongly suggest that the true average thickness of such lenses is different from desired value.

Problem:9

A chemical process has produced, on the average, 800 tons of chemical per day. The daily yields for the past week are 785, 805, 790, 793, and 802 tons. Do these data indicate that the average yield is less than 800 tons and hence that something is wrong with the process? Test at the 5% level of significance. What assumptions must be satisfied in order for the procedure that you used to analyze these data to be valid?

Solution:

x	$x-\bar{x}$	$(x-\bar{x})^2$
785	-10	100
805	10	100
790	-5	25
793	-2	4
802	-7	49
$\sum x = 3975 \quad \bar{x} = 795$		$\sum (x-\bar{x})^2 = 278$ $\frac{1}{n-1}\sum (x-\bar{x})^2 = 69.5$

Solution:

Here, n = 5, μ=800;

H0: μ≥800 & H1:μ<800

$s = \sqrt{69.5} = 8.34$

Here it is one tailed t test

to validly use a one-sample t-test:

1. The daily yields are independent observations.
2. The population distribution of yields is approximately normal

The level of significance is 0.05.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{795 - 800}{8.34/\sqrt{5}} = -5 / 3.73 = -1.34$$

∴calculated t value = -1.34

The critical value from table with 4 degrees of freedom and 0.05 LOS is -2.132

Since the calculated value is greater than the critical value, we accept null hypothesis.

Conclusion: At the 5% significance level, the data do not provide sufficient evidence that the average daily output has fallen below 800 tons.

Problem:10

Is there any systematic tendency for part-time college faculty to hold their students to different standards than do full-time faculty? The article “Are There Instructional Differences Between Full-Time and Part-Time Faculty?”(*College Teaching,* 2009: 23–26) reported that for a sample of 125 courses taught by full-time faculty, the mean course GPA was 2.7186 and the standard deviation was .63342, whereas for a sample of 88 courses taught by part-timers, the mean and standard deviation were 2.8639 and .49241, respectively. Does it appear that true average course GPA for part-time faculty differs from that for faculty teaching full-time? Test the appropriate hypotheses at significance level .01 by first obtaining a *P*-value.

Solution:

Given Full-time faculty

$$n_1 = 125$$
$$\bar{x}_1 = 2.7186$$
$$s_1 = 0.63342$$

Part-time faculty

$$n_2 = 88$$
$$\bar{x}_2 = 2.8639$$
$$s_2 = 0.49241$$

$$H_0: \mu_1 = \mu_2$$
$$H_a: \mu_1 \neq \mu_2$$

Two-tailed test, $\alpha = 0.01$.

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$z = \frac{2.8639 - 2.7186}{\sqrt{\frac{0.63342^2}{125} + \frac{0.49241^2}{88}}}$$

$$= \frac{0.1453}{\sqrt{\frac{0.4012}{125} + \frac{0.24247}{88}}} = \frac{0.1453}{\sqrt{0.0032096 + 0.0027556}}$$

$$= \frac{0.1453}{\sqrt{0.0059652}} = \frac{0.1453}{0.0772} = 1.882$$

P-value

Two-tailed test:

$$p = 2P(Z > 1.882) = 2(0.0305) = 0.061$$

0.061>0.01⇒Fail to reject H_0

Conclusion: At the 0.01 significance level, there is no significant evidence that part-time and full-time faculty differ in their average course GPA.

Problem:11

Persons having Reynaud's syndrome are apt to suffer a sudden impairment of blood circulation in fingers and toes. In an experiment to study the extent of this impairment, each subject immersed a forefinger in water and the resulting heat output(cal/cm^2 /min)was measured. **For m=10** subjects with the syndrome, the **average heat output was** $\bar{x} = \mathbf{0.64}$, and for **n=10** non sufferers, the true average output was 2.05.Let μ_1 and μ_2 denote the true average heat outputs for the two types of subjects. Assume that the two distributions of heat output are normal with $\sigma_1 = 0.2$ and $\sigma_2 = 0.4$.

Consider testing H_0: $\mu_1 - \mu_2$=-1.0 versus H_a: : $\mu_1 - \mu_2$ <-1.0 at level 0.01. Describe in words what H_a says, and then carry out the test.

Work Integrated Learning Programmes

Solution:

Given m=10 ;$\bar{x}_1 = 0.64; \sigma_1 = 0.20; n = 10$;$\bar{x}_2 = 2.05; \sigma_2 = 0.40$

$$H_0: \mu_1 - \mu_2 = -1.0$$
$$H_a: \mu_1 - \mu_2 < -1.0$$

Two-sample z-test:

$$z = \frac{\left(\bar{x}_1 - \bar{x}_2\right) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}}$$

$$\bar{x}_1 - \bar{x}_2 = 0.64 - 2.05 = -1.41$$

$$z = \frac{-1.41 - (-1.0)}{\sqrt{\frac{0.2^2}{10} + \frac{0.4^2}{10}}}$$

$$z = \frac{-0.41}{\sqrt{\frac{0.04}{10} + \frac{0.16}{10}}} = \frac{-0.41}{\sqrt{0.004 + 0.016}}$$

$$= \frac{-0.41}{\sqrt{0.020}} = \frac{-0.41}{0.1414} = -2.90$$

Critical Value: Left-tailed test at $\alpha = 0.01$:$z_{0.01} = -2.33$

Since $-2.90 < -2.33$ **Reject** H_0**.**

At the 0.01 significance level, there is strong evidence that the true mean heat output of Reynaud's syndrome sufferers is more than 1.0 unit lower than that of non-sufferers.

This supports the claim that people with Reynaud's syndrome experience substantially greater impairment of blood circulation than the null hypothesis assumes.

Problem:12

Two catalysts may be used in a batch chemical process. Twelve batches were prepared using catalyst 1, resulting in an average yield of 86 and a sample standard deviation of 3.Fifteen batches were prepared using catalyst 2, and they resulted in an average yield of 89 with a standard deviation of 2. Assume that yield measurements are approximately normally distributed with the same standard deviation. Is there evidence to support a claim that catalyst 2 produces a higher mean yield than catalyst 1? Use $\alpha = 0.01$.

Solution:

Catalyst 1

$$n_1 = 12$$
$$\bar{x}_1 = 86$$
$$s_1 = 3$$

Catalyst 2

$$n_2 = 15$$
$$\bar{x}_2 = 89$$
$$s_2 = 2$$

The population standard deviations are assumed to be **equal**.
Significance level: $\alpha = 0.01$

H_0: $\mu_1 \geq \mu_2$; H_a: $\mu_2 > \mu_1 \Rightarrow \mu_1 < \mu_2$ left tailed test.

Pooled Variance

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$
$$s_p^2 = \frac{(11)(3^2) + (14)(2^2)}{25}$$
$$s_p^2 = \frac{99 + 56}{25} = \frac{155}{25} = 6.2$$
$$s_p = \sqrt{6.2} = 2.49$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$t = \frac{86 - 89}{2.49\sqrt{\frac{1}{12} + \frac{1}{15}}} = \frac{-3}{2.49\sqrt{0.0833 + 0.0667}}$$

$$t = \frac{-3}{2.49\sqrt{0.15}}$$

$$t = \frac{3}{2.49 \times 0.3873} = \frac{3}{0.9643} = -3.11$$

df=n_1+n_2−2=25

Critical Value at $\alpha = 0.01$(left-tailed)

$$t_{0.01,25} = -2.485 \text{(approximately)}$$

Since

$$t = -3.11 < -2.485$$

Reject H_0.

At the 0.01 significance level, there is strong evidence that catalyst 2 produces a higher mean yield than catalyst 1.

Problem: 13

A random sample of 150 recent donations at a certain blood bank reveals that 82 were type A blood. Does this suggest that the actual percentage of type A donations differs from 40%, the percentage of the population having type A blood? Carry out a test of the appropriate hypotheses using a significance level of .01. Would your conclusion have been different if a significance level of .05 had been used?

Solution:

Given

Sample size: $n = 150$

Number of type A donations: $x = 82$

Sample proportion:

$$\hat{p} = \frac{82}{150} = 0.5467$$

Claimed population proportion:

$$p_0 = 0.40$$

$$H_0: p = 0.40$$

$$H_a: p \neq 0.40$$

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

$$z = \frac{0.5467 - 0.40}{\sqrt{0.40 \times 0.60/150}}$$

$$z = \frac{0.1467}{\sqrt{0.24/150}} = \frac{0.1467}{\sqrt{0.0016}} = \frac{0.1467}{0.04} = 3.6675$$

for a two-tailed test at significance level 0.01:

$$z_{\alpha/2} = z_{0.005} = 2.576$$

Rejection region:

$$|\, z \,| > 2.576$$

Since:

$$3.6675 > 2.576$$

Reject H_0.

There is strong evidence that the true percentage of type A blood donations differs from 40%.

Would the conclusion change at $\alpha = 0.05$?

At $\alpha = 0.05$:

$$z_{0.025} = 1.96$$

Because:

$$3.6675 > 1.96$$

Reject H_0.

No, the conclusion would not change.

At both significance levels (0.01 and 0.05), we reject the claim that the proportion of type A donations is 40%.

Problem:14

A company manufactured a product on two machines i.e. Machine A and Machine B. Consider the following information related to sampling of these to test and compare them. Based on this information, can we conclude that the product manufactured by Machine A is superior to Machine B.

Machine	Sample size	Mean	Standard Deviation
Machine A	15	5.5	0.5
Machine B	15	5.1	0.2

Solution:

Given $\bar{x}$= 5.5, $\bar{y}$=5.1, s_1=0.5, s_2=0.2, n_1=n_2=15

The pooled standard deviation. $Sp^2=\frac{((n1-1)(s1)^2+(n2-1)(s2)^2}{(n1+n2-2)}=$

$$\frac{14(0.25)+14(0.04)}{28}=\frac{4.06}{28}=0.145$$

To test

H0: The product manufactured by Machine A is not superior to Machine B

H1: The product manufactured by Machine A is superior to Machine B

The test statistic = t= $(\bar{x}-\bar{y})/\sqrt{sp^2.(\frac{1}{n_1}+\frac{1}{n_2})}=\frac{5.5-5.1}{\sqrt{0.145(\frac{1}{15}+\frac{1}{15})}}=\frac{0.4}{0.139}=2.8776$

P-value=P(t(28df)< 2.8776) <0.00001 which is <0.01 and 0.05 levels of significance, (or calculated t =2.8776 > the critical t(28df)= 2.4679at 1% los and 1.701 at 5% los) for one tailed (right tailed) test therefore, Ho is rejected and H1 is accepted.

Problem:15

An urban community would like to show that the incidence of breast cancer is higher in their area than in a nearby rural area. (PCB levels were found to be higher in the soil of the urban community.) If it is found that 20 of 200 adult women in the urban community have breast cancer and 10 of 150 adult women in the rural community have breast cancer, can we conclude at the 0.05 level of significance that breast cancer is more prevalent in the urban community?

Solution:

Given:
Urban community: $n_1 = 200$,$x_1 = 20$
Rural community: $n_2 = 150$,$x_2 = 10$
Significance level: $\alpha = 0.05$

$$H_0: p_1 = p_2 (breastcancerrateisthesame)$$

$$H_a: p_1 > p_2 \text{ (urban rate is higher)}$$

$$\hat{p}_1 = \frac{20}{200} = 0.10$$

$$\hat{p}_2 = \frac{10}{150} \approx 0.0667$$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{20 + 10}{200 + 150} = \frac{30}{350} = 0.0857$$

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$z = \frac{0.10 - 0.0667}{\sqrt{0.0857(1-0.0857)(1/200 + 1/150)}} = 1.101$$

Critical value:
$z_{0.05}$=1.645

Since the calculated value is less than the critical value, we accept null hypothesis and reject alternate hypothesis.

Therefore, there is **not sufficient evidence** to conclude that the incidence of breast cancer is higher in the urban community compared with the rural community.

Problem:16

An agency conducted a survey to understand the trend related to student's preferences in taking admissions

Program	Number of Students
B.E(CSE)	320
B.E(AIML)	240
B.E(DSE)	350
B.E(Cyber Security)	200
B.E(IoT)	90
Total	1200

Is it reasonable to conclude that there is no preference among the five B E programs offered as mentioned above?
(Level of significance = 0.05)

Solution:

H_0: The student preference is independent of the program

H_1: The student preference is not independent of the program

Program	Number of Students (O)	Students Expected (E)	$\frac{(E-O)^2}{E}$
B.E(CSE)	320	$\frac{1200}{5}=240$	26.66667
B.E(AIML)	240	$\frac{1200}{5}=240$	0
B.E(DSE)	350	$\frac{1200}{5}=240$	50.41667
B.E(Cyber Security)	200	$\frac{1200}{5}=240$	6.66667
B.E(IoT)	90	$\frac{1200}{5}=240$	93.75
Total	**1200**		**177.5**

Then $\chi^2 = 177.5$

The critical value from table with LOS = 0.05 and 4 degrees of freedom is 9.488

Hence H0 is rejected

Conclusion: The student preference is not independent of program

Problem:17

BITS Pilani University researcher wants to determine whether gender (Male or Female) is independent of preference for a new product (Like or Dislike). A random sample of 250 people is surveyed, and the results are recorded in the contingency table below:

Test at a 5% significance level whether gender and preference for the product are independent.

Gender	Like	Dislike	Total
Male	80	40	120
Female	90	40	130
Total	170	80	250

Solution:

Step 1: Define the Hypotheses

Null Hypothesis (H0): Gender and preference for the new product are **independent**.

Alternative Hypothesis (H1): Gender and preference for the new product are **not independent**.

Step 2: Compute Expected Frequencies

The expected frequency (E) for each cell in a contingency table is calculated as:

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

For each cell:

$$E_{Male,Like} = \frac{120 \times 170}{250} = \frac{20400}{250} = 81.6$$

$$E_{Male,Dislike} = \frac{120 \times 80}{250} = \frac{9600}{250} = 38.4$$

$$E_{Female,Like} = \frac{130 \times 170}{250} = \frac{22100}{250} = 88.4$$

$$E_{Female,Dislike} = \frac{130 \times 80}{250} = \frac{10400}{250} = 41.6$$

Gender	Like (Observed)	Like (Expected)	Dislike (Observed)	Dislike (Expected)
Male	80	81.6	40	38.4
Female	90	88.4	40	41.6

Solution contd.,

Step 3: Compute the Chi-square Test Statistic

The formula for **Chi-square** is:

$$\chi 2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 = \frac{(80-81.6)^2}{81.6} + \frac{(40-38.4)^2}{38.4} + \frac{(90-88.4)^2}{88.4} + \frac{(40-41.6)^2}{41.6}$$

$$= \frac{(-1.6)^2}{81.6} + \frac{(1.6)^2}{38.4} + \frac{(1.6)^2}{88.4} + \frac{(-1.6)^2}{41.6}$$

$$= \frac{2.56}{81.6} + \frac{2.56}{38.4} + \frac{2.56}{88.4} + \frac{2.56}{41.6}$$

$$= 0.0314 + 0.0667 + 0.0290 + 0.0615 = 0.1886$$

Step 4: Find the Critical Value

Degrees of freedom (df) =

= (2 - 1) × (2 - 1) = 1

Significance level (α) = **0.05**

From the Chi-square table,

the critical value for df=1, at alpha = 0.05 is **3.841**.

Step 5: Compare & Make a Decision

$\chi^2 = 0.1886$ **is less than 3.841, we fail to reject H0.**

Step 6: Conclusion

At the **5% significance level**, there is **no significant evidence** to suggest that gender and preference for the new product are related.

This means gender and product preference **are independent**.

Problem:

In a study conducted in the Forestry and Wildlife Department at Virginia Tech, J. A. Wesson examined the influence of the drug succinylcholine on the circulation levels of androgens in the blood. Blood samples were taken from wild, free-ranging deer immediately after they had received an intramuscular injection of succinylcholine administered using darts and a capture gun. A second blood sample was obtained from each deer 30 minutes after the first sample, after which the deer was released. The levels of androgens at time of capture and 30 minutes later, measured in nanograms per milliliter (ng/mL), for 15 deer are given in Table 10.2. Assuming that the populations of androgen levels at time of injection and 30 minutes later are normally distributed, test at the 0.05 level of significance whether the androgen concentrations are altered after 30 minutes.

Table 10.2: Data for Case Study 10.1

	Androgen (ng/mL)		
Deer	At Time of Injection	30 Minutes after Injection	d_i
1	2.76	7.02	4.26
2	5.18	3.10	−2.08
3	2.68	5.44	2.76
4	3.05	3.99	0.94
5	4.10	5.21	1.11
6	7.05	10.26	3.21
7	6.60	13.91	7.31
8	4.79	18.53	13.74
9	7.39	7.91	0.52
10	7.30	4.85	−2.45
11	11.78	11.10	−0.68
12	3.90	3.74	−0.16
13	26.00	94.03	68.03
14	67.48	94.03	26.55
15	17.04	41.70	24.66

Solution: Let μ_1 and μ_2 be the average androgen concentration **at the time of injection** and **30 minutes after injection**, respectively.

Deer	At Time of Injection	30 Minutes After Injection	(d_i)
1	2.76	7.02	4.26
2	5.18	3.10	−2.08
3	2.68	5.44	2.76
4	3.05	3.99	0.94
5	4.10	5.21	1.11
6	7.05	10.26	3.21
7	6.60	13.91	7.31
8	4.79	18.53	13.74
9	7.39	7.91	0.52
10	7.30	4.85	−2.45
11	11.78	11.10	−0.68
12	3.90	3.74	−0.16
13	26.00	94.03	68.03
14	67.48	94.03	26.55
15	17.04	41.70	24.66

Hypotheses

$$H_0:\ \mu_1 = \mu_2\ or\, \mu_D = \mu_1 - \mu_2 = 0$$
$$H_1:\ \mu_1 \neq \mu_2\ or\, \mu_D = \mu_1 - \mu_2 \neq 0$$

α=0.05

we use a **paired t-test**.
Let d_i =androgen level 30 minutes after injection − androgen level at time of injection.
The test statistic is

$$t = \frac{\bar{d} - \Delta_0}{s_d/\sqrt{n}}$$

where $\Delta_0 = 0$and $n = 15$.

Degrees of freedom: $\nu = n - 1 = 14$

$$t_{0.025,14} = 2.145$$

$$\bar{d} = 9.848$$

$$s_d{}^2 = \frac{1}{n-1}\sum(d-\bar{d})^2 = \frac{4777.85}{14} = 341.25$$

$$s_d = 18.474$$

Thus,

$$t = \frac{9.848 - 0}{18.474/\sqrt{15}} = 2.06$$

Calculated t value is less than critical value.
Hence we accept the null hypothesis

Deer	X	Y	(d_i)	d-$\bar{d}$	(d−$\bar{d}$)^2
1	2.76	7.02	4.26	-5.588	31.22574
2	5.18	3.10	−2.08	-11.928	142.2772
3	2.68	5.44	2.76	-7.088	50.23974
4	3.05	3.99	0.94	-8.908	79.35246
5	4.10	5.21	1.11	-8.738	76.35264
6	7.05	10.26	3.21	-6.638	44.06304
7	6.60	13.91	7.31	-2.538	6.441444
8	4.79	18.53	13.74	3.892	15.14766
9	7.39	7.91	0.52	-9.328	87.01158
10	7.30	4.85	−2.45	-12.298	151.2408
11	11.78	11.10	−0.68	-10.528	110.8388
12	3.90	3.74	−0.16	-10.008	100.1601
13	26.00	94.03	68.03	58.182	3385.145
14	67.48	94.03	26.55	16.702	278.9568
15	17.04	41.70	24.66	14.812	219.3953

Thank You

Work Integrated Learning Programmes